AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 31 · WORKBOOK

Design the Decision Process

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Design the Decision Process before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your **explanation**.

Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Generate, Retrieve, or Calculate?

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Design the Decision Process trace and record the first step you can explain without help.

Use W2: Track the current **candidate** for Designing a complete decision **process**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Design the Decision Process explanation against the supplied facts before discussing it.

Use the worked case for Designing a complete decision **process**. Proposed **explanation**: “One failed row proves no match”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Design a complete room decision process

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Design the Decision Process target to a trace and a justified conclusion.

Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Design the Decision Process case and show the deciding evidence.

A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

Trace glass and repair the rule.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Write a decision **process**: a clear pencil goes to the pencil tray, a clear book to the shelf, and an unknown object to review. Trace a lunch box.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 31 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Design the Decision Process adds to Helpful Classroom Sorter and keep the evidence.

Complete the sorter v1 **decision flow** and annotate every **branch** with the **evidence** it uses.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

My independently reasoned answer

Record support used: none, hint or teacher **explanation**.

WORKBOOK · 12 OF 21

W1 Describe all possible kinds of ending

> Describe all possible kinds of ending. Apply the supplied Design the Decision Process facts and keep your reasoning.

Name the three kinds of final **output** in the activity **process** and the condition that leads to each. Explain why a **candidate** failing its time check is not automatically a final no-match result.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Track the current candidate

> Track the current candidate. Apply the supplied Design the Decision Process facts and keep your reasoning.

Trace 9 minutes with Quiet required No. Record the current **candidate**, time result, quiet condition and next step until the **process** stops. Explain why the request values should not change as the **candidate** advances.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair an incorrect branch

> Repair an incorrect branch. Apply the supplied Design the Decision Process facts and keep your reasoning.

A design says “If the activity is quiet OR its time fits, suggest it.” For 5 minutes with Quiet required Yes it selects B1. Identify the failed requirement and replace the condition with a correct one.

Give the correct result for this request.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Design a complete room decision process

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Design a complete room decision process. Apply the supplied Design the Decision Process facts and keep your reasoning.

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

> **My reasoning and evidence**

WORKBOOK · 16 OF 21

W5 Preserve the first compatible option

> Preserve the first compatible option. Apply the supplied Design the Decision Process facts and keep your reasoning.

For the activity catalogue, 15 minutes with Quiet required No allows several activities. Explain why returning the last compatible row breaks the agreed rule even though it fits.

Describe a stopping rule that preserves the first match, and one way to check that a loop cannot repeat forever.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Design the Decision Process

> A fictional worksheet routes craft kits by required glue and number of scissors. Explain the labelled step before checking the changed condition.

A fictional worksheet routes craft kits by required glue and number of scissors.

K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4.

Request: glue required Yes, scissors needed 2. Return first kit meeting both requirements.

AI ACADEMY / AI-1/31

Design the Decision Process

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate fields

Require Yes/No glue input and whole scissors count 0-4.



2. Evaluate K1

Scissors fit, but required glue is absent: continue.



3. Evaluate K2

Glue is present and 2 scissors meet the request: accept.



4. Stop

Return K2 without replacing it with later K3.

CHANGE ONE THING

Increase scissors needed to 3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Where does the first original rejection occur, and why is K3 not originally selected?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional worksheet routes craft kits by required glue and number of scissors.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return K2 without replacing it with later K3.

1. Stop

2. Evaluate K1

3. Evaluate K2

4. Validate fields

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: Stop Return K2 without replacing it with later K3. A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase scissors needed to 3. Which conclusion is supported by the supplied facts?

1. K3 is selected.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: K3 is selected. K1 lacks glue;

> K2 has only 2 scissors; K3 has glue and 4.

Repair a missing building block

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same **explanation**?.

Return to this week and explain what you changed.

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two need separate approval?. Return to this week and explain what you changed.

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

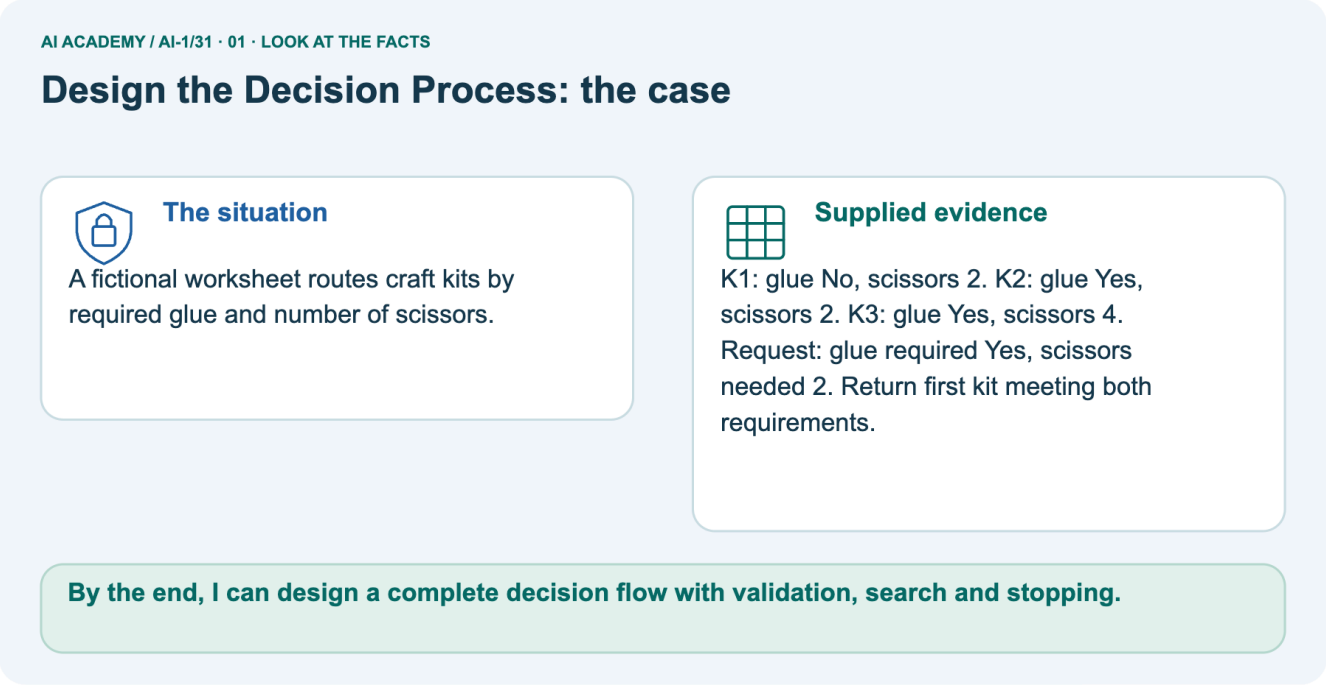
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Design the Decision Process picture from its labels, then record a relevant limit.

01. Design the Decision Process: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

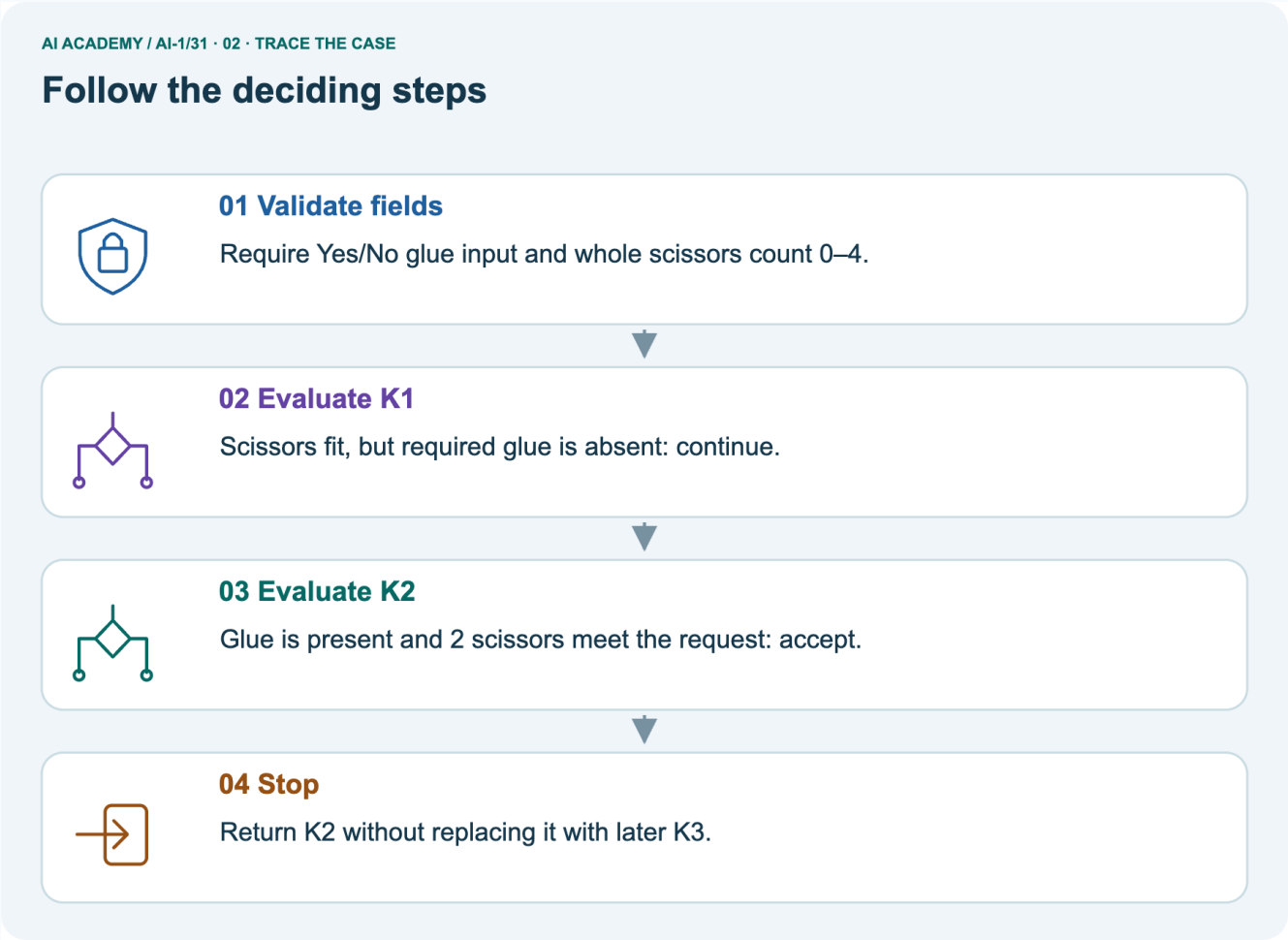
A fictional worksheet routes craft kits by required glue and number of scissors. Supplied facts: K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4. Request: glue required Yes, scissors needed 2.

Return first kit meeting both requirements.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



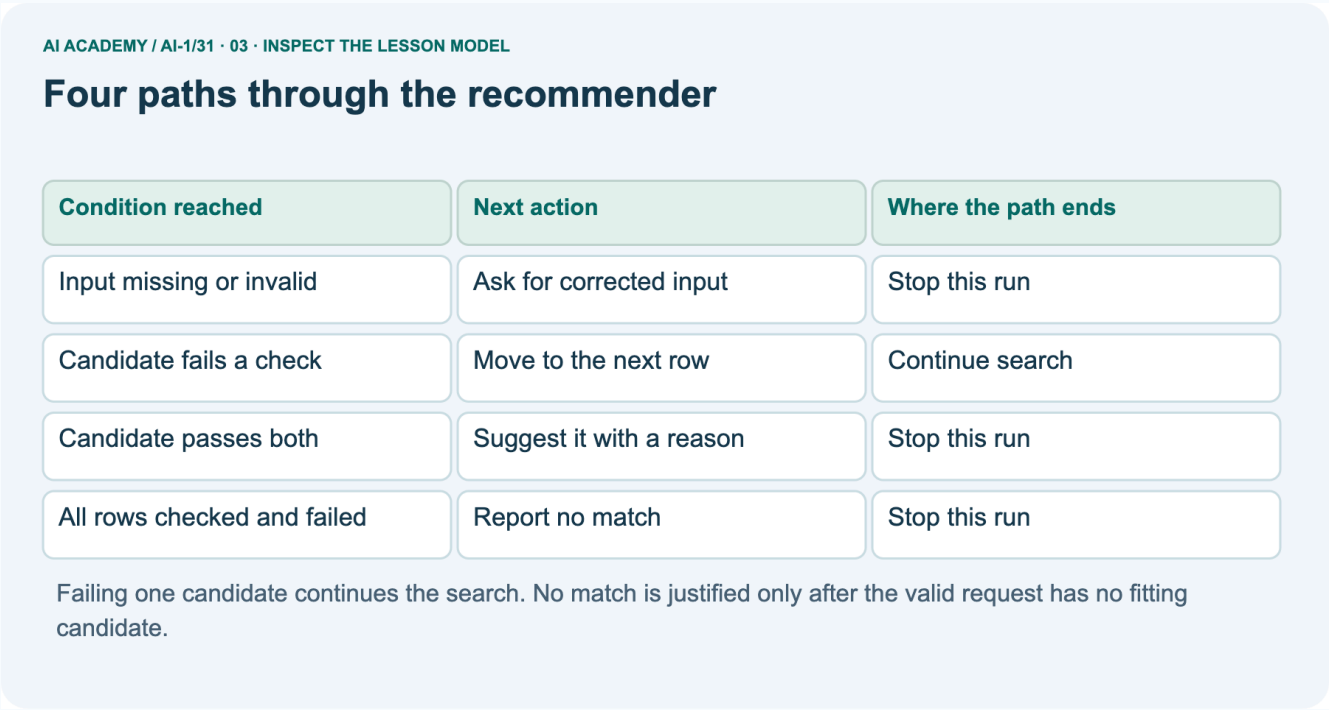
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate fields: Require Yes/No glue **input** and whole scissors count 0–4. → Evaluate K1: Scissors fit, but required glue is absent: continue. → Evaluate K2: Glue is present and 2 scissors meet the request: accept. → Stop: Return K2 without replacing it with later K3.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Four paths through the recommender



Follow the labelled picture with the matching explanation. Open the vector diagram

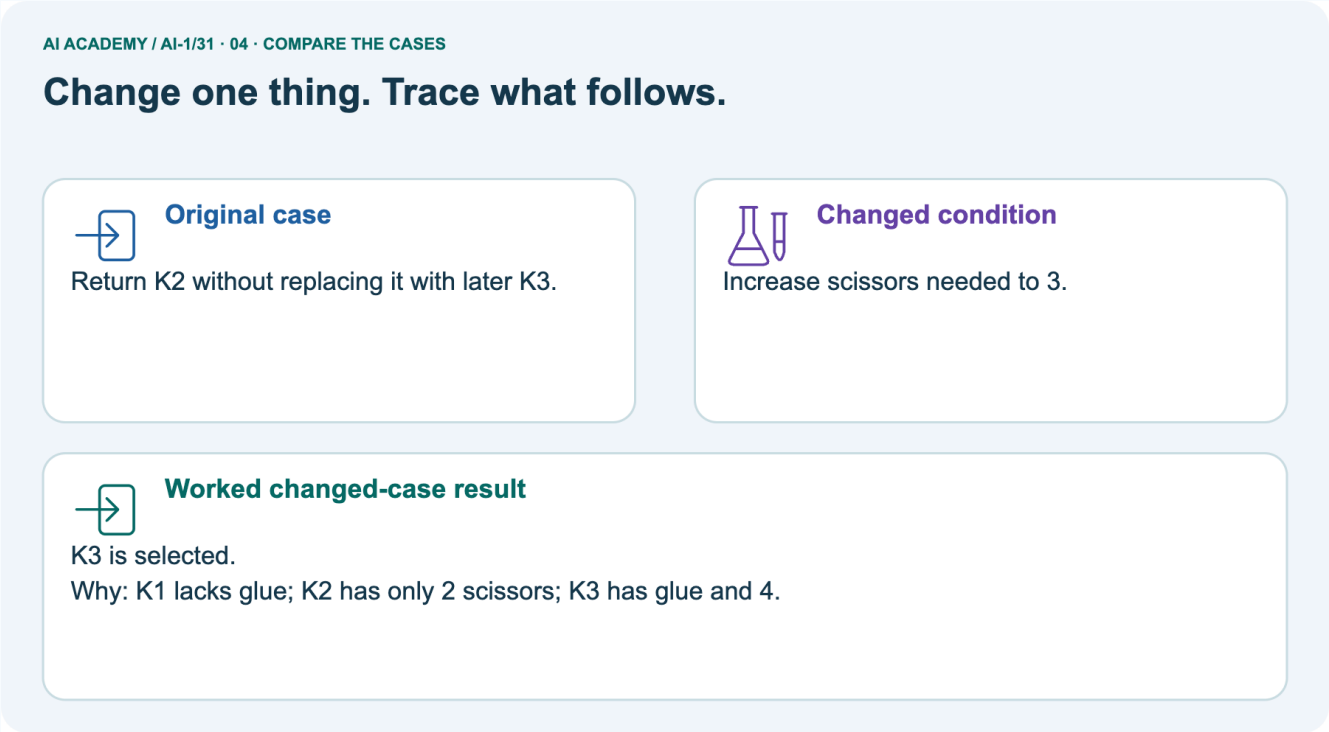
Four paths through the recommender. **Input** missing or invalid | Ask for corrected **input** | Stop this run. **Candidate** fails a check | Move to the next row | Continue search.

Candidate passes both | Suggest it with a reason | Stop this run. All rows checked and failed | Report no match | Stop this run

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



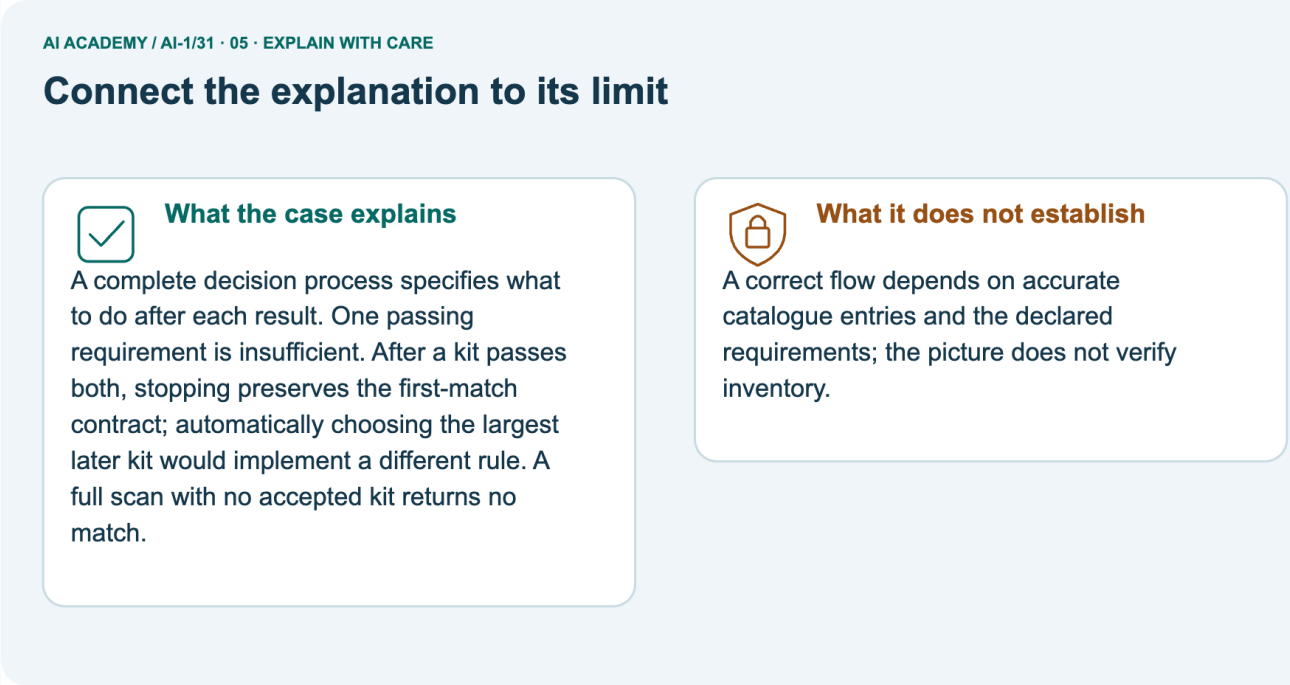
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return K2 without replacing it with later K3. Changed condition: Increase scissors needed to 3. Expected result: K3 is selected. Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

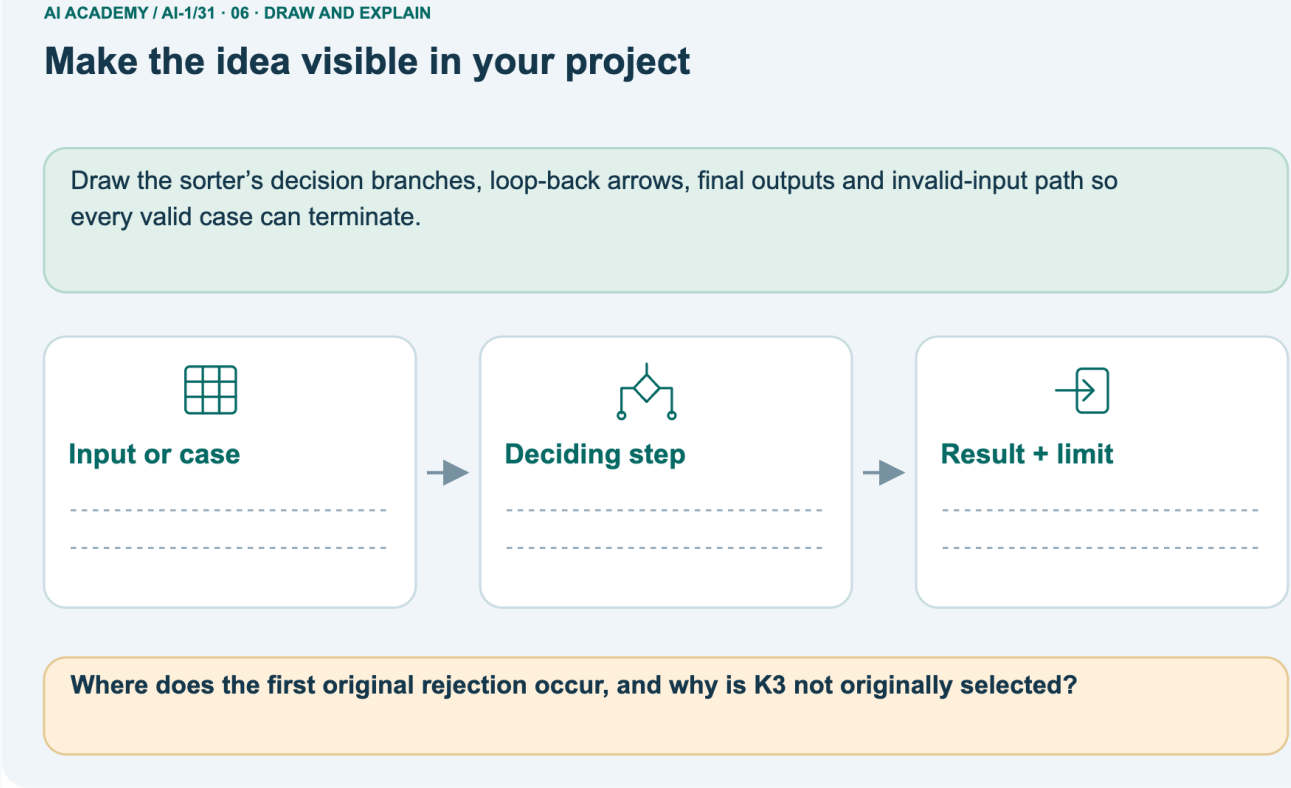
After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Limit: A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate. Draw your **input**, deciding step, result and limit.

Supported question: Where does the first original rejection occur, and why is K3 not originally selected?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.